
Exercise sheet 4 for Numerical Optimization

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Exercise 1 (Characterization of convexity on behalf of the first derivative, 5+4 points)

Let $C \subset \mathbb{R}^n$ be open and convex and $f : C \rightarrow \mathbb{R}$ continuously differentiable. Prove that

- a) f is convex on $C \Leftrightarrow$ for all $x, y \in C$ it holds true that

$$f(x) - f(y) \geq \nabla f(y)^T (x - y) .$$

Hint: For $\Rightarrow$ consider $\lim_{\lambda \rightarrow 0} \frac{f(\lambda x + (1-\lambda)y) - f(y)}{\lambda}$ and for $\Leftarrow$ consider $z = \lambda x + (1-\lambda)y$ and show that $\lambda(f(x) - f(z)) + (1-\lambda)(f(y) - f(z)) \geq 0$.

- b) f is strictly convex on $C \Leftrightarrow$ for all $x, y \in C$ with $x \neq y$ it holds true that

$$f(x) - f(y) > \nabla f(y)^T (x - y) .$$

Hint: For $\Rightarrow$: „ $\geq$ “ follows from a). Why cannot „ $>$ “ be proven considering $\lim_{\lambda \rightarrow 0} \frac{f(\lambda x + (1-\lambda)y) - f(y)}{\lambda}$? Instead, assume „ $\geq$ “ and consider the point $z = \frac{x+y}{2}$.

Exercise 2 ((Strict/uniform) convexity of monomials, 3+3+4+4 points)

Let $C \subset \mathbb{R}^n$ with $C \neq \emptyset$ be convex. A function $f : C \rightarrow \mathbb{R}$ is called *uniformly convex* in C if there is a constant $\mu > 0$ such that for all $x, y \in C$ and $\lambda \in [0, 1]$ it holds true that

$$f(\lambda x + (1 - \lambda)y) + \mu\lambda(1 - \lambda)\|x - y\|_2^2 \leq \lambda f(x) + (1 - \lambda)f(y) . \quad (1)$$

- a) What is the illustrative difference between convexity, strict convexity and uniform convexity? Why do the implications *uniform convexity* $\Rightarrow$ *strict convexity* $\Rightarrow$ *convexity* hold true?

Consider $p_n : \mathbb{R} \rightarrow \mathbb{R}$, $p_n(x) = x^n$, the monomial of order $n \in \mathbb{N}_0 = \{0, 1, 2, 3, \dots\}$.

- b) Prove that p_n is convex on $\mathbb{R}$ if and only if $n = 0$, $n = 1$ or $n \geq 2$ and even.
- c) Prove that p_n is strictly convex on $\mathbb{R}$ if and only if $n \geq 2$ and even.
- d) Prove that p_n is uniformly convex on $\mathbb{R}$ if and only if $n = 2$.

Exercise 3 (Convex optimization problems, 3+3+3+3+1+1+1+1+1+1+1+1+1+1+1 points)

A convex optimization problem is defined as

$$\text{find } \hat{x} \in \mathbb{R}^n \text{ such that } f(\hat{x}) = \min_{x \in C} f(x) \quad (2)$$

with convex domain $C \subset \mathbb{R}^n$ and convex function $f : C \rightarrow \mathbb{R}$. We defined convexity of sets and functions in order to define convex optimization problems. These are fundamental for numerical optimization due to the following statements. Prove these statements.

- a) Every local minimum of f in C is a global minimum.
- b) If $C \subset \mathbb{R}^n$ is open and $f \in C^1(C)$, every stationary point of f in C is a global minimum.
- c) The set of solutions of (2) is convex (but could be empty).
- d) If f is strictly convex in C , the convex optimization problem (2) has at most one solution.

In order to get a better intuition for these kind of problems, we now try to find simple, one-dimensional example problems.

- e) Draw a convex, but not strictly convex, function that has exactly one local minimum.
- f) Draw a convex, but not strictly convex, function that does not have a local minimum.
- g) Draw a convex function that has more than one local minimum.
- h) Try to draw a convex function having a local minimum that is not a global minimum and explain why this is not possible.
- i) Try to draw a convex function having a local maximum and explain why this is not possible.
- j) Try to draw a convex function having a saddle point and explain why this is not possible.
- k) Try to draw a convex function having a non-convex set of solutions of (2) and explain why this is not possible.
- l) Draw a strictly convex function that has exactly one local minimum.
- m) Draw a strictly convex function that does not have a local minimum.
- n) Try to draw a strictly convex function that has more than one local minimum and explain why this is not possible.

Exercise 4 (Direct search methods - Hooke-Jeeves method, 4+1 points)

Over the next weeks, we will be looking at unconstrained optimization, i.e., optimization problems without constraints. First we will have a look at *direct search methods* respectively *derivative-free methods*. These methods do not require any derivatives or regularities. They rely exclusively on evaluations of the objective function.

The idea is as follows. These methods can be used for functions that are not differentiable or whose derivatives are difficult to compute. However, one has to accept that in general there are no convergence statements and even if these methods converge, they usually perform worse than methods requiring more conditions.

Direct search methods are highly relevant for practical applications. Imagine objective functions that are complex black boxes, then it is difficult to approximate derivative values with acceptable accuracy and effort.

The first direct search method is the *Hooke-Jeeves method*. For $f : \mathbb{R}^n \rightarrow \mathbb{R}$, the idea is to try to find its minimizer by successively searching along all coordinate axes. In each iteration step, an *exploration step* is performed to find a direction in which the function decreases comparatively rapidly. In the subsequent *progression step*, the next iterate is determined based on the direction from the exploration step.

Exploration step: Consider a current iterate $x^{(k)} \in \mathbb{R}^n$ and a vector of step sizes $h \in \mathbb{R}_+^n$, where $h_j, j = 1, \dots, n$, is the step size along the coordinate axis of the j -th dimension. Then, one after the other, the directions of the coordinate axes of all n dimensions are examined. For the j -th dimension it is checked whether

$$f(x^{(k)} + h_j e_j) < f(x^{(k)}) ,$$

where e_j is the j -th unit vector of $\mathbb{R}^n$. If so, then $h_j e_j$ is added to the iterate. If not, then $x^{(k)} - h_j e_j$ is examined and if

$$f(x^{(k)} - h_j e_j) < f(x^{(k)}) ,$$

then $-h_j e_j$ is added to the iterate. After having tested all n dimensions, we obtain

$$y^{(k)} \quad \text{with} \quad y_j^{(k)} \leftarrow x_j^{(k)} + \begin{cases} h_j e_j & \text{first check successful} \\ -h_j e_j & \text{first check not successful, second check successful} \\ 0 & \text{both checks not successful} \end{cases}$$

and it holds true that

$$f(y^{(k)}) \leq f(x^{(k)}) .$$

If $y^{(k)} = x^{(k)}$, the exploration step did not produce progress, the step size is set to half and a new exploration step is carried out. If $y^{(k)} \neq x^{(k)}$, a direction of decrease is found and the subsequent progression step is carried out.

Progression step: The direction of decrease is $y^{(k)} - x^{(k)}$, so you could actually search along the straight line

$$x^{(k)} + t(y^{(k)} - x^{(k)}) , \quad t > 0$$

for the lowest function value and select this point as the next iterate $x^{(k+1)}$. But how can one do this line search without having any derivatives of the objective function available? For this reason, the line search is usually replaced by setting $t = 2$, which results in

$$w^{(k)} \leftarrow x^{(k)} + 2(y^{(k)} - x^{(k)}) = 2y^{(k)} - x^{(k)} ,$$

and then performing a further exploration step starting from $w^{(k)}$. We denote the result of this second exploration step as $z^{(k)}$. If the second exploration step yields an improvement, so $f(z^{(k)}) < f(y^{(k)})$, the new iterate is set to $z^{(k)}$. If not, it is set to $y^{(k)}$.

The complete algorithm can be found in the lecture notes.

- a) We want to implement the Hooke-Jeeves method. To do this, fill in the gaps in the MATLAB programme `hooke_jeeves.m`.
- b) The MATLAB programme `test_hooke_jeeves.m` tests your implementation of the Hooke-Jeeves method on both Rosenbrock's and Himmelblau's function. Execute it and reflect on the results.